

Prompt Engineering Cheat Sheet - Terms to Retain (Lessons 1, 2, & 3)

Core Concepts

- Prompt: The input you give to an AI model to guide its response.
- Prompt Engineering: The skill of crafting effective, precise prompts to get high-quality AI outputs.
- Text-to-Text Generative AI: AI that takes text as input and generates text as output (e.g., ChatGPT, Claude, Gemini, LLaMA-2).

Basic Prompt Elements

- Role Prompting: Assigning a role to the AI (e.g., "You are a project manager.")
- Instruction-Based Prompting: Giving direct tasks to the AI.
- Context: Background information included in the prompt to make AI responses relevant.
- Audience: Who the AI is writing for (e.g., students, executives, customers).
- Purpose: The goal of the output (e.g., inform, persuade, summarize).
- Constraints: Rules such as word limits, style, tone, or format restrictions.
- Format Instructions: Telling AI to output in a specific structure (list, table, bullet points, etc.).

Prompting Techniques

- Few-Shot Prompting: Giving the AI a few input/output examples to mimic.
- Chain-of-Thought Prompting: Asking AI to think step by step before answering.
- Salty Prompt: Adding style and tone instructions to make AI's output feel more natural, playful, formal, etc.
- Exploratory Prompting: Open-ended prompting used for brainstorming or idea generation.
- Structured Prompting: Prompting for repeatable, predictable formats (lists, bullet points, templates).
- Conversational Prompting: Creating dialogue-style interactions (used for chatbots, roleplays, tutoring).

- Functional Prompting: For direct task completion (translate, summarize, generate code, etc.).

Output Control Terms

- Output Control: Controlling how the AI structures the response.
- Length Control: Limiting output with word count, sentence count, or character limits.
- Step-by-Step Reasoning: Making AI show its thought process explicitly.
- Markdown Table: A table written using Markdown syntax (common in AI outputs).
- JSON Output: Requesting AI to output structured data in JSON format.

Tone & Style Control Terms

- Tone Keywords: Words like formal, casual, empathetic, persuasive, playful used to shape style.
- Stacking: Combining role + tone + format + length in a single prompt.

Quality Assurance Terms

- Evaluation: Checking if the AI's response meets the goal (clarity, accuracy, style, completeness).
- Prompt-Based Self-Evaluation: Asking the AI to critique or improve its own output.
- Iteration: Adjusting the prompt gradually until the output is satisfactory.

Common AI Issues

- Hallucination: When AI generates information that is false or not grounded in reality.
- Grounding: Forcing the AI to stick to a given source or input text only.
- Verification Prompt: Asking the AI to check its own information before answering.